

COMSATS University Islamabad, Vehari Campus

Information Communication Technology (CSC101)

Lab 11

Topic: Conditional Structure in Python

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Objective

Learn how to use conditional statements to control program flow.

1. Useful Concepts

1.1 If Statement

Definition: Executes code only when the condition is TRUE.

Syntax:

```
if(condition):  
    # code
```

Example:

```
x = 4  
if(x == 4):  
    print("Correct value")
```

Output:

Correct value

1.2 If-Else Statement

Definition: Executes one block if condition is true, otherwise another block.

Syntax:

```
if(condition):  
    # code
```

```
else:  
    # code
```

Example:

```
x = 3  
  
if(x == 4):  
    print("Correct")  
else:  
    print("Wrong")
```

Output:

Wrong

1.3 If-Elif-Else Statement

Definition: Used to check multiple conditions.

Syntax:

```
if(condition1):  
    # code  
  
elif(condition2):  
    # code  
  
else:  
    # code
```

Example:

```
x = 2  
  
if(x == 1):  
    print("One")  
elif(x == 2):  
    print("Two")  
else:  
    print("Other")
```

Output:

Two

1.4 Nested If Statement

Definition: An if statement inside another if statement.

Syntax:

```
if(condition1):  
    if(condition2):  
        # code
```

Example:

```
x = 2  
y = 10  
  
if(x < 5):  
    if(y < 20):  
        print("Both conditions are true")
```

Output:

Both conditions are true

2. Solved Lab Activities

Activity 1: If Statement

Python program to illustrate an if statement:

Code:

```
x = 4  
  
if x == 3:  
    print('Lions are the king of the jungle')  
  
if x == 4:  
    print('Bears eat honey')
```

Output:

Bears eat honey

Activity 2: If-Else

Python program to illustrate an if else statement:

Code:

```
x = 4
if x == 3:
    print('Lions are the king of the jungle')
else:
    print('Bears eat honey')
```

Output:

Bears eat honey

Activity 3: If-Elif

Python program to illustrate a function definition with return:

Code:

```
x = 4
if x == 3:
    print('Lions are the king of the jungle')
elif x == 4:
    print('Canberra is the capital of Australia')
else:
    print('Bears eat honey')
```

Output:

Canberra is the capital of Australia

Activity 4: Nested If

Python program to illustrate nested if else statements:

Code:

```
x = 1
y = 11

if x <= 2 and y < 20:
    print('The numbers x and y fall under the criteria')
    sum_val = x + y
    if sum_val < 50:
        print('The sum of x and y is:', sum_val)
else:
    print('The numbers x and y dont fall under the criteria')
```

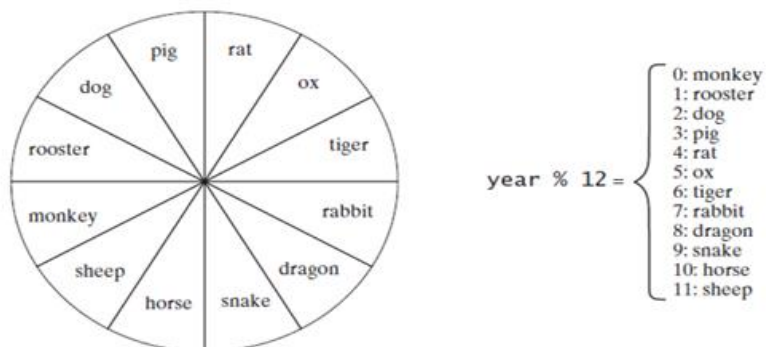
Output:

The numbers x and y fall under the criteria

The sum of x and y is: 12

Activity 5: Chinese Zodiac

Write a program to find out the Chinese zodiac sign for a given year. The Chinese zodiac sign is based on a 12-year cycle, and each year in this cycle is represented by an animal—monkey, rooster, dog, pig, rat, ox, tiger, rabbit, dragon, snake, horse, and sheep:



Code:

```
year = int(input("Enter a year: "))
month = int(input("Enter a month (1-12): "))

zodiacYear = year % 12

if zodiacYear == 0:
    print("Monkey")
elif zodiacYear == 1:
    print("Rooster")
elif zodiacYear == 2:
    print("Dog")
elif zodiacYear == 3:
    print("Pig")
elif zodiacYear == 4:
    print("Rat")
elif zodiacYear == 5:
    print("Ox")
elif zodiacYear == 6:
    print("Tiger")
elif zodiacYear == 7:
    print("Rabbit")
elif zodiacYear == 8:
    print("Dragon")
elif zodiacYear == 9:
    print("Snake")
elif zodiacYear == 10:
    print("Horse")
else:
    print("Sheep")
```

Output Example:

Enter a year: 1963

Enter a month (1-12): 5

Rabbit

3. Graded Lab Tasks

Task 1: Write a program to check whether an integer is positive, negative, or zero.

Code:

```
num = 5
if num > 0:
    print("Positive")
elif num < 0:
    print("Negative")
else:
    print("Zero")
```

Output:

Positive

Task 2:

Write a program to input marks of five subjects Physics, Chemistry, Biology, Mathematics, and Computer. Calculate percentage and grade according to following:

Percentage $\geq$ 90% : Grade A

Percentage $\geq$ 80% : Grade B

Percentage $\geq$ 70% : Grade C

Percentage $\geq$ 60% : Grade D

Percentage $\geq$ 40% : Grade E

Percentage $<$ 40% : Grade F

Code:

```
physics = 85
chemistry = 80
biology = 75
mathematics = 90
computer = 88
percentage = (physics + chemistry + biology + mathematics + computer)
/ 5
if percentage >= 90:
    grade = 'A'
elif percentage >= 80:
    grade = 'B'
elif percentage >= 70:
    grade = 'C'
elif percentage >= 60:
    grade = 'D'
elif percentage >= 40:
    grade = 'E'
else:
    grade = 'F'
print("Percentage:", percentage)
print("Grade:", grade)
```

Output:

Percentage: 83.6

Grade: B

Task 3: Write a program to check whether the triangle is equilateral, isosceles or scalene triangle.

Code:

```
a = 5
b = 5
c = 5

if a == b == c:
    print("Equilateral")
elif a == b or b == c or a == c:
    print("Isosceles")
else:
    print("Scalene")
```

Output:

Equilateral

Task 4:

Write a dummy authentication system program in which you accept user inputs for email and password. Let's say the correct email and password are abc@gmail.com and abc respectively. If the email and password entered are correct it should display "User is logged in". If the email is correct, then prompt the user that the password is not correct. If the password is correct then prompt the user to enter the correct email. If both are incorrect then display the corresponding message.

Code:

```
email = 'abc@gmail.com'
password = 'abc'

if email == 'abc@gmail.com' and password == 'abc':
    print("User is logged in")
elif email == 'abc@gmail.com':
    print("Incorrect password")
elif password == 'abc':
    print("Incorrect email")
else:
    print("Both wrong")
```

Output:

User is logged in

Task 5: Write a program to check whether a year is a leap year or not.**Code:**

```
year = 2024

if (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0):
    print("Leap year")
else:
    print("Not leap year")
```

Output:

Leap year

✓ End of Lab